EXPERIMENT 7

Aim: Implementation of any one Graph Traversal Technique for rea1-world application

Program:

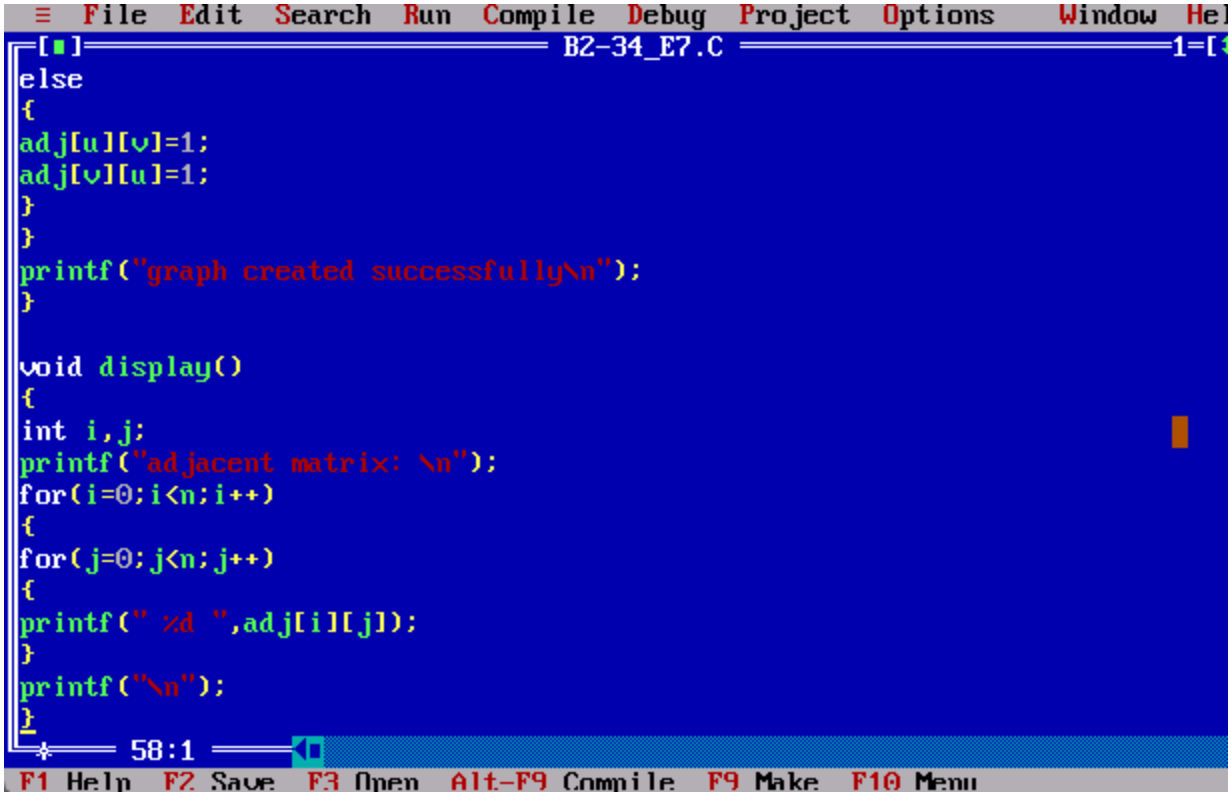
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                        B2-34_E7.C                              1=[
#include <stdio.h>
#include <conio.h>

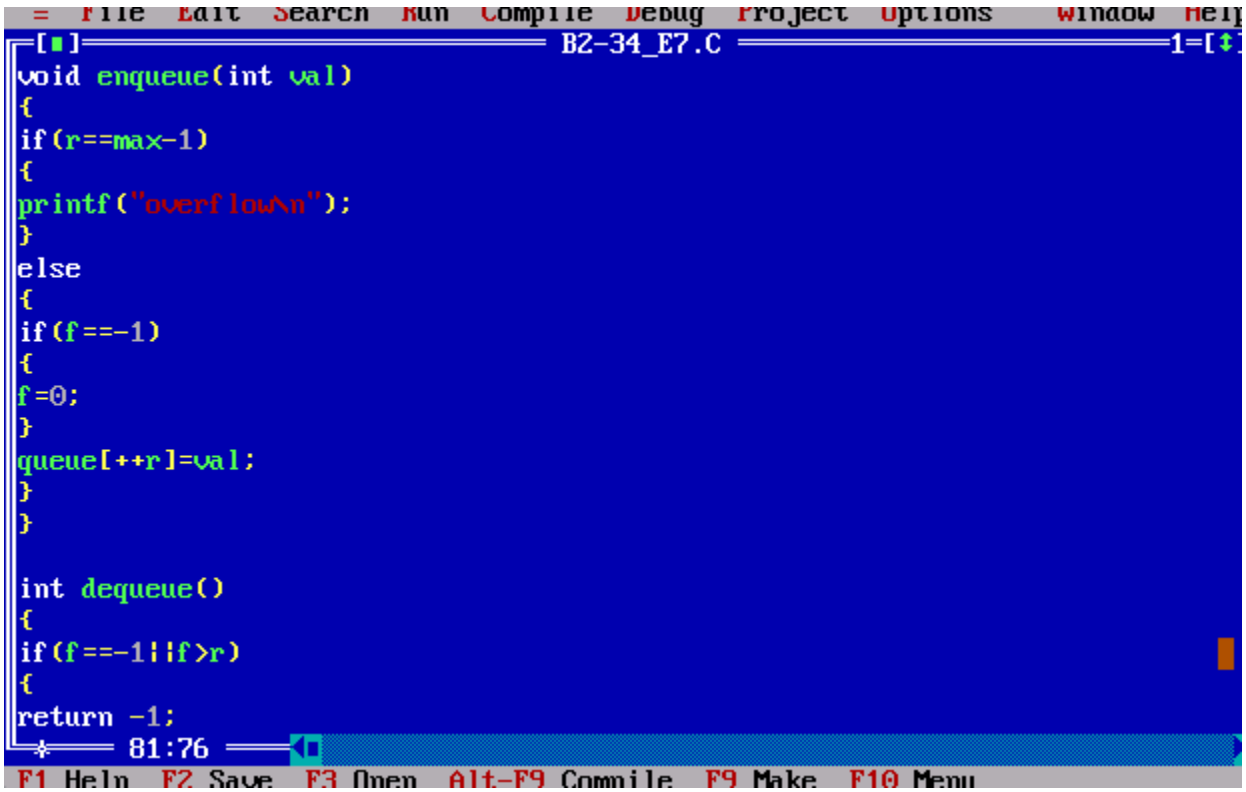
#define max 100

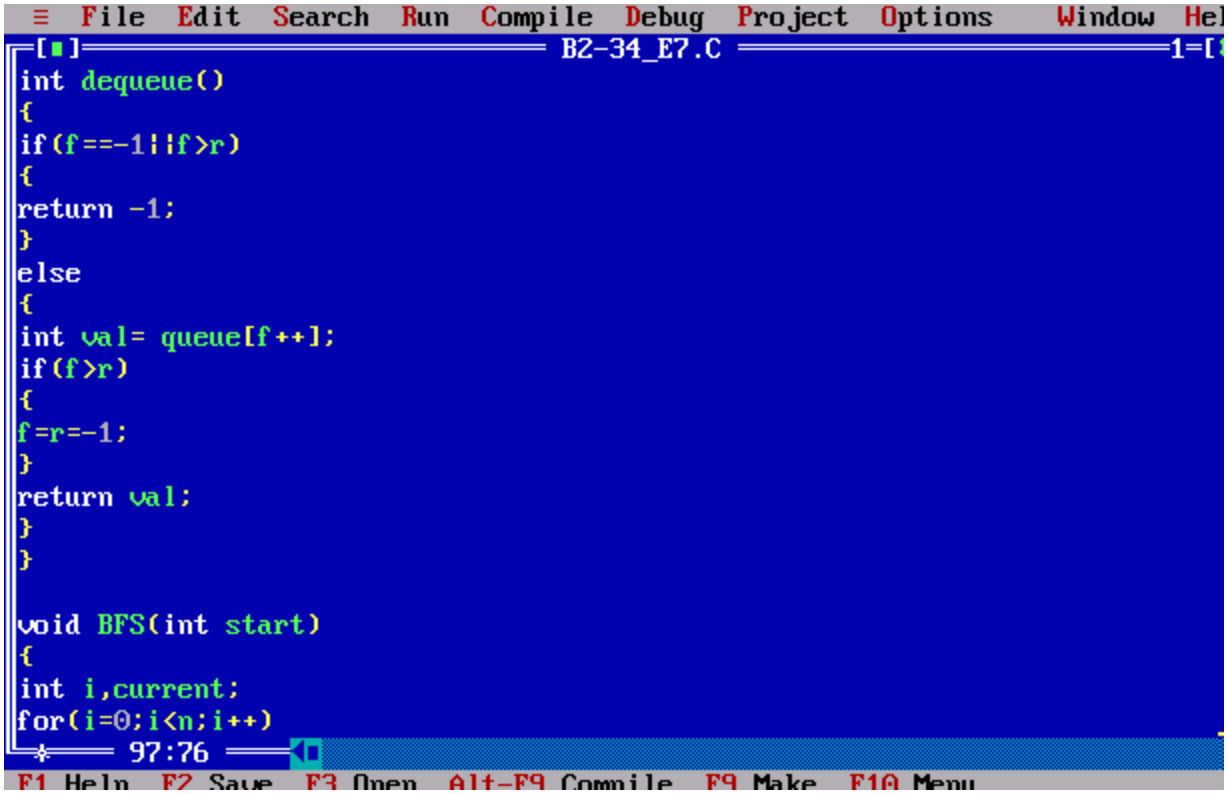
int adj[max][max],visited[max],n;
int queue[max],f=-1,r=-1;

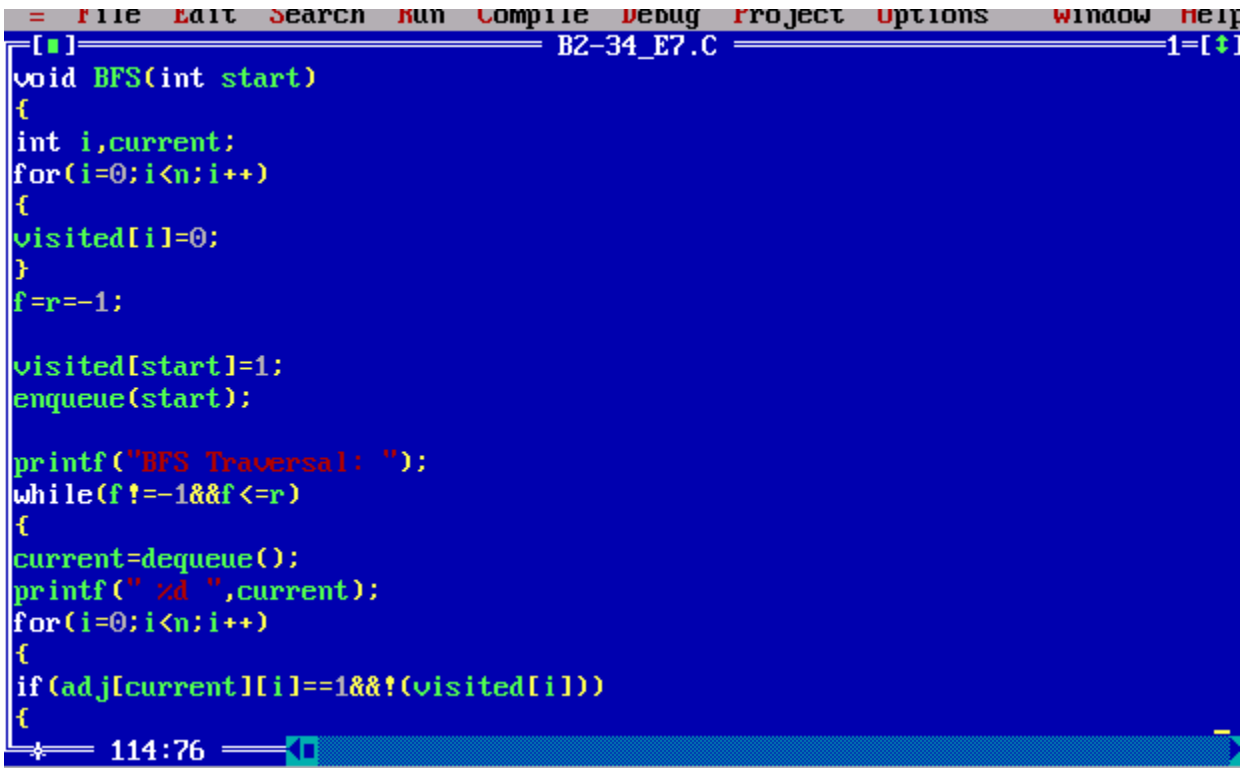
void creategraph()
{
int i,j,u,v,edges;
printf("enter no of vertices: ");
scanf("%d",&n);
if(n<0||n>max)
{
printf("invalid no of vertices\n");
}
else
{
for(i=0;i<n;i++)
{
```

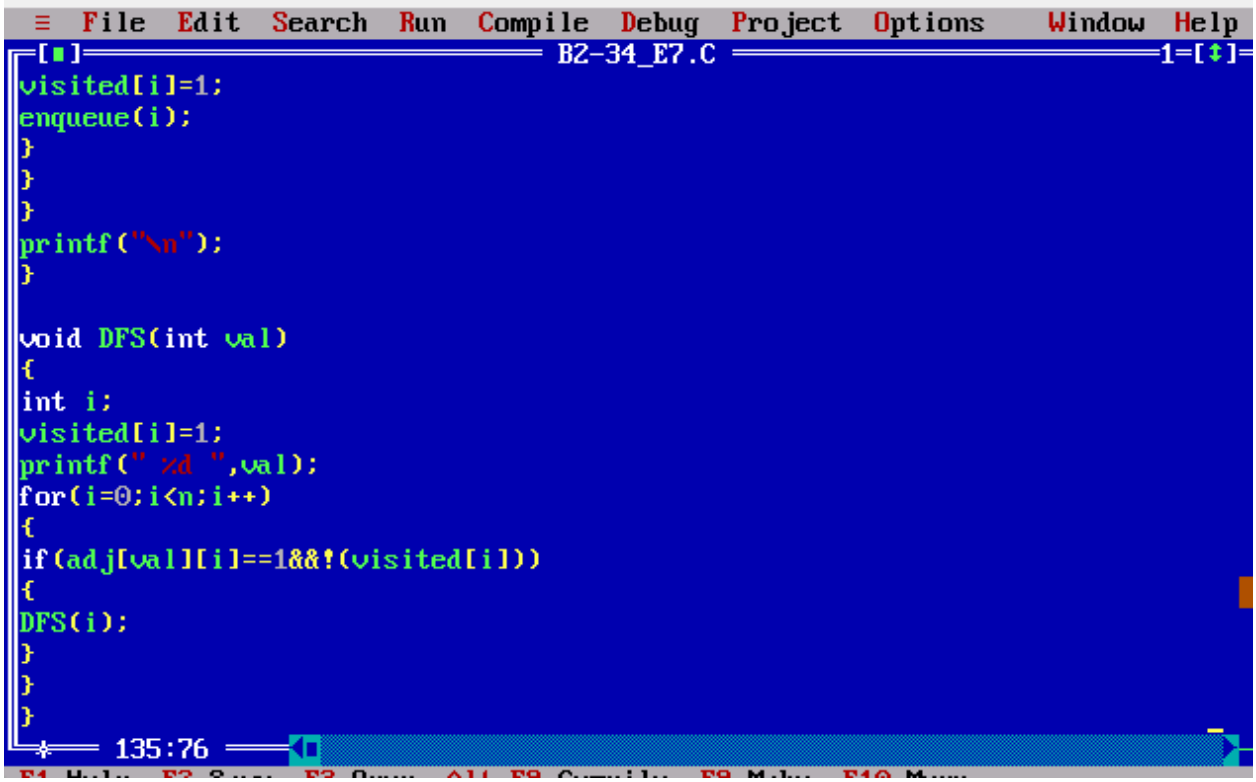
```
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                        B2-34_E7.C                              1=[
for(i=0;i<n;i++)
{
for(j=0;j<n;j++)
{
adj[i][j]=0;
}
}
}
printf("enter no of edges: ");
scanf("%d",&edges);
for(i=0;i<edges;i++)
{
printf("enter vertices u,v: ");
scanf("%d %d",&u,&v);
if(u<0||v<0||u>=n||v>=n)
{
printf("invalid vertices\n");
}
else
{
adj[u][v]=1;
```

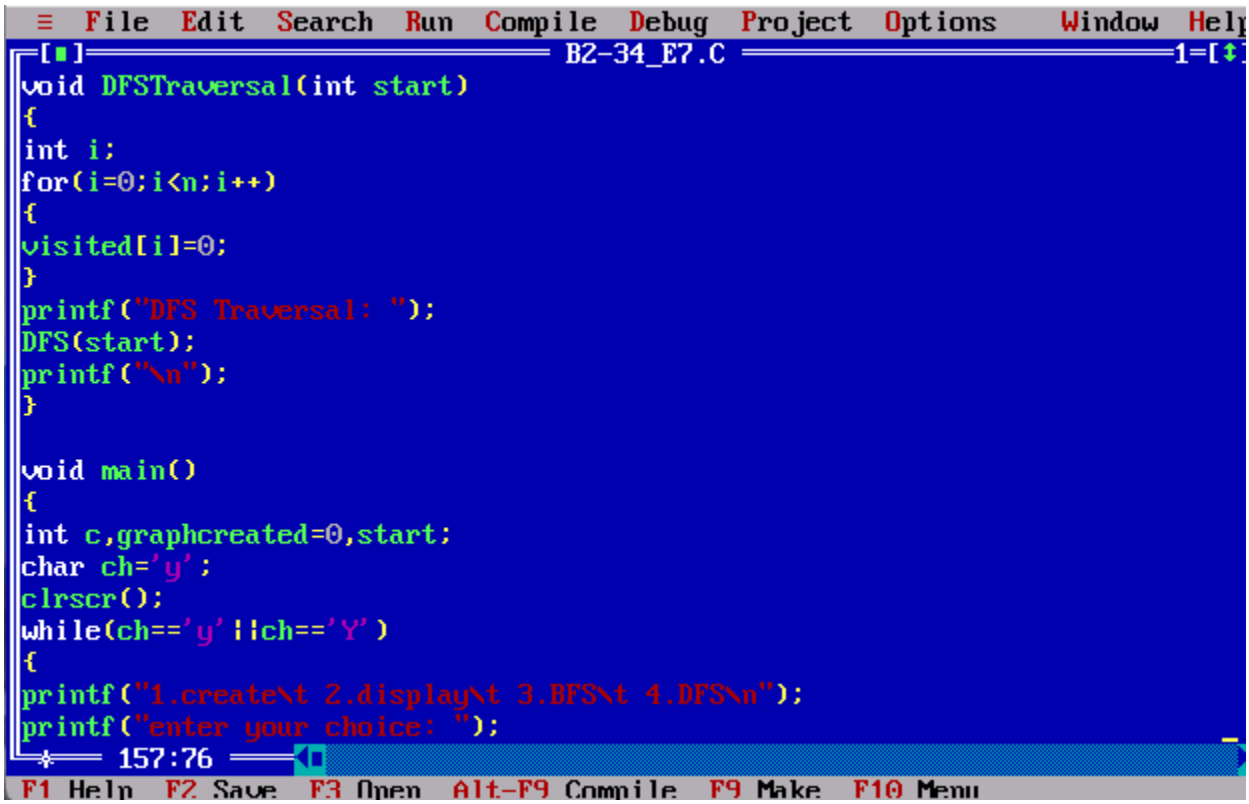

File Edit Search Run Compile Debug Project Options Window
B2-34_E7.C
else
{
adj[u][v]=1;
adj[v][u]=1;
}
}
printf("graph created successfully\n");
}
void display()
{
int i,j;
printf("adjacent matrix: \n");
for(i=0;i<n;i++)
{
for(j=0;j<n;j++)
{
printf(" %d ",adj[i][j]);
}
printf("\n");
}
58:1
F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu

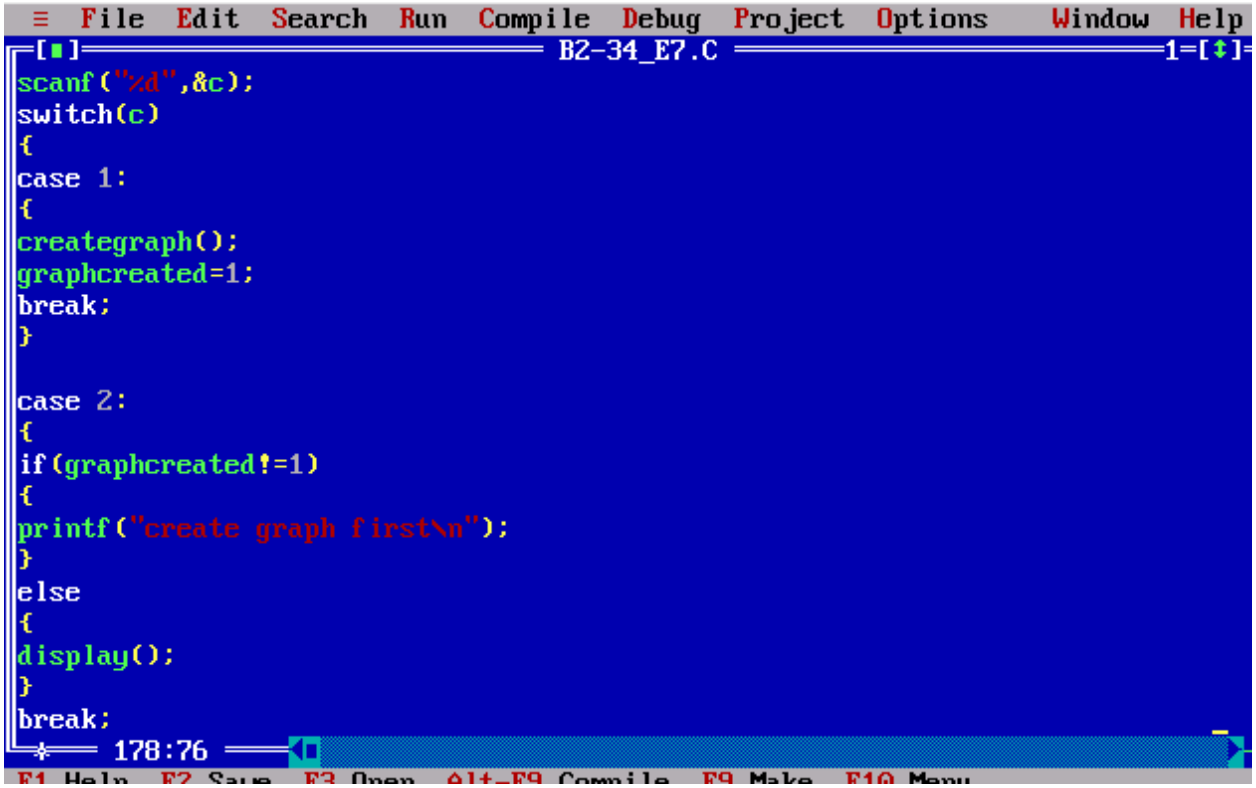

B2-34_E7.C
void enqueue(int val)
{
if(r==max-1)
{
printf("overflow\n");
}
else
{
if(f==-1)
{
f=0;
}
queue[++r]=val;
}
}
int dequeue()
{
if(f==-1||f>r)
{
return -1;
81:76
F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu

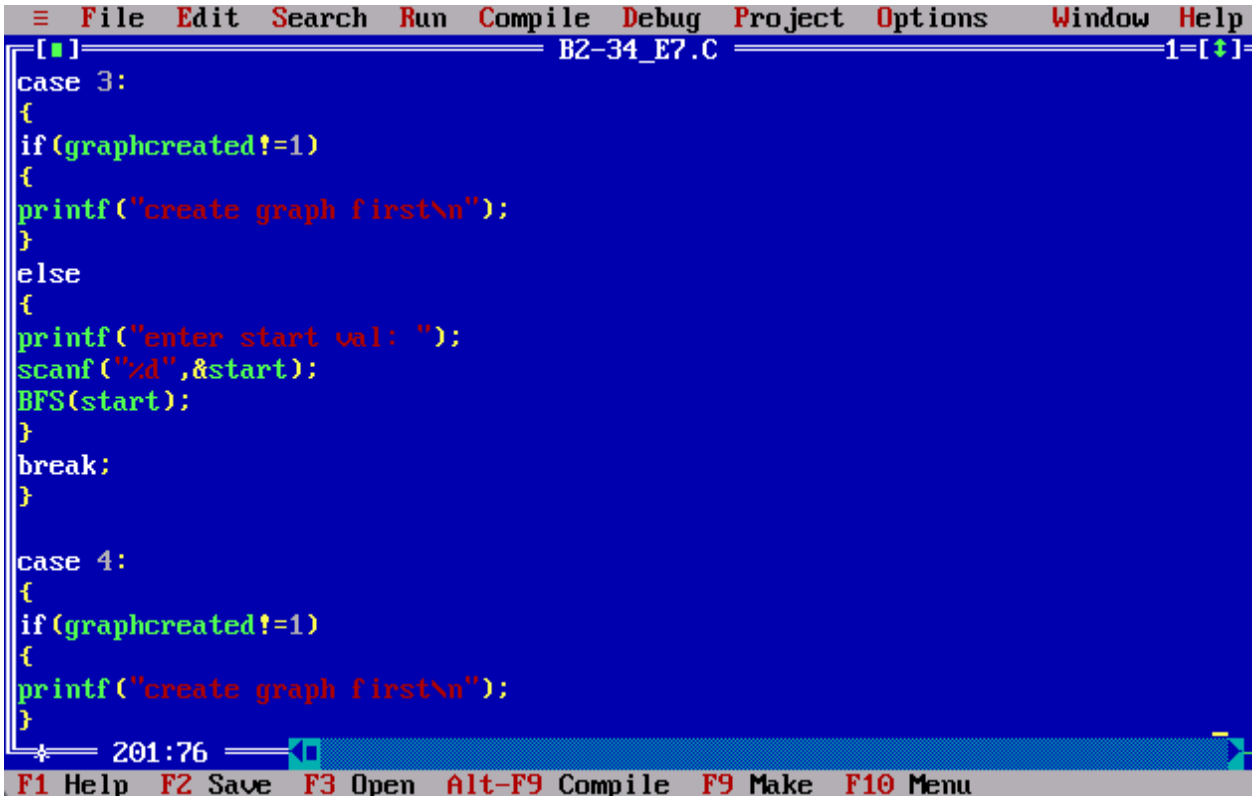

File Edit Search Run Compile Debug Project Options Window
B2-34_E7.C
int dequeue()
{
if(f==-1||f>r)
{
return -1;
}
else
{
int val= queue[f++];
if(f>r)
{
f=r=-1;
}
return val;
}
}
void BFS(int start)
{
int i,current;
for(i=0;i<n;i++)
97:76
F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu


File Edit Search Run Compile Debug Project Options Window
B2-34_E7.C
void BFS(int start)
{
int i,current;
for(i=0;i<n;i++)
{
visited[i]=0;
}
f=r=-1;
visited[start]=1;
enqueue(start);
printf("BFS Traversal: ");
while(f!=-1&&f<=r)
{
current=dequeue();
printf(" %d ",current);
for(i=0;i<n;i++)
{
if(adj[current][i]==1&&!(visited[i]))
{
114:76


File Edit Search Run Compile Debug Project Options Window Help
B2-34_E7.C
visited[i]=1;
enqueue(i);
}
}
}
printf("\n");
}
void DFS(int val)
{
int i;
visited[i]=1;
printf(" %d ",val);
for(i=0;i<n;i++)
{
if(adj[val][i]==1&&!(visited[i]))
{
DFS(i);
}
}
}
135:76


File Edit Search Run Compile Debug Project Options Window Help
B2-34_E7.C
void DFSTraversal(int start)
{
int i;
for(i=0;i<n;i++)
{
visited[i]=0;
}
printf("DFS Traversal: ");
DFS(start);
printf("\n");
}
void main()
{
int c,graphcreated=0,start;
char ch='y';
clrscr();
while(ch=='y'||ch=='Y')
{
printf("1.create\t 2.display\t 3.BFS\t 4.DFS\n");
printf("enter your choice: ");
157:76
F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu


File Edit Search Run Compile Debug Project Options Window Help
B2-34_E7.C
scanf("%d",&c);
switch(c)
{
case 1:
{
creategraph();
graphcreated=1;
break;
}
case 2:
{
if(graphcreated!=1)
{
printf("create graph first\n");
}
else
{
display();
}
break;
178:76


File Edit Search Run Compile Debug Project Options Window Help
B2-34_E7.C
case 3:
{
if(graphcreated!=1)
{
printf("create graph first\n");
}
else
{
printf("enter start val: ");
scanf("%d",&start);
BFS(start);
}
break;
}
case 4:
{
if(graphcreated!=1)
{
printf("create graph first\n");
}
201:76
F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu

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[■]═══════════════ B2-34_E7.C ═══════════════1=[↕]
}
else
{
printf("enter start val: ");
scanf("%d",&start);
DFSTraversal(start);
}
break;
}

default:
{
printf("invalid choice\n");
break;
}
}
printf("do you want to continue?(y/n): ");
scanf(" %c",&ch);
}
getch();
}
═══ 221:76 ═══
```

```
1.create        2.display       3.BFS   4.DFS
enter your choice: 1
enter no of vertices: 3
enter no of edges: 3
enter vertices u,v: 0 1
enter vertices u,v: 1 2
enter vertices u,v: 0 2
graph created successfully
do you want to continue?(y/n): Y
1.create        2.display       3.BFS   4.DFS
enter your choice: 2
adjacent matrix:
 0  1  1
 1  0  1
 1  1  0
do you want to continue?(y/n): y
1.create        2.display       3.BFS   4.DFS
enter your choice: 3
enter start val: 2
BFS Traversal:  2  0  1
do you want to continue?(y/n):
```